Predicting Race Finish Times at 100% Effort from Personal Strava Data

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Seminar Big Data Methods for Economists
Final Project

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Outline

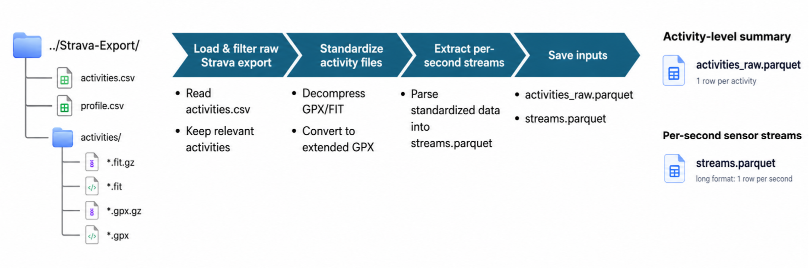
Motivation

- Running is one of the most accessible forms of endurance sport.
- It is associated with major physical and mental health benefits.
- At the same time, running has become increasingly data-driven:
  - GPS watches record distance, pace, elevation and heart rate.
  - Platforms such as Strava turn everyday training into structured data.
- This creates a practical question: can personal training data be used to predict race performance?

Research Question

Given only an athlete's open Strava export, how accurately can a transparent statistical or machine-learning model predict race finish times?

Turning Raw Strava Exports into Modelling-Ready Data



Activities Data Overview

- **Feature matrix**
  - Final feature table: **495 rows** and **21 features**.

Group	Variables
Volume	`distance_km`, `moving_s`, `mean_gap_speed_mps`
Heart rate	`mean_hr`, `p50_hr`, `p90_hr`
Training load	`atl`, `ctl`, `tsb`
Effort / calendar	`effort`, `dow`, `month`

- **Grade-Adjusted Pace (GAP)**
  - Converts observed speed to a flat-course equivalent by correcting for local slope:

$$\mathrm{GAP} = v \cdot (1 + 3.3g + 32.5g^2), \qquad g = \frac{\Delta\mathrm{elevation}}{\Delta\mathrm{distance}}$$

  - v = instantaneous speed; g = GPS-derived gradient, clipped to reduce noise.

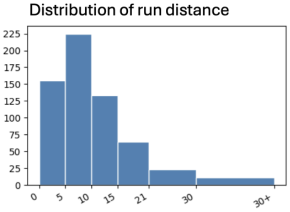
Data Quality and Sample Weighting

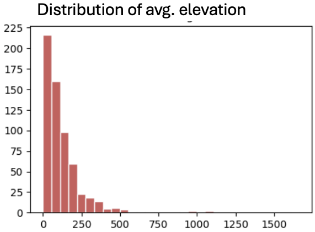
Each training observation is assigned a confidence weight $w \in [0, 1]$:

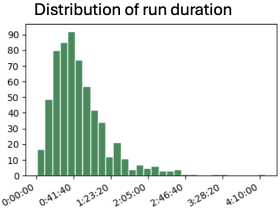
$$w = \underbrace{c_{\text{GPS}}}_{\text{valid speed}} \times \underbrace{c_{\text{ele}}}_{\text{valid elevation}} \times \underbrace{c_{\text{HR}}}_{\text{valid HR}} \times \underbrace{c_{\text{dur}}}_{\geq 30\,\text{min}}$$

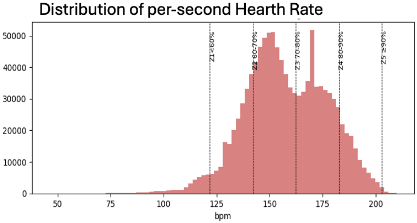
- Each c_i = fraction of per-second records with a valid sensor reading (0 = none, 1 = complete).
- c_{dur} ramps linearly from 0 to 1 for moving time below 30 min (very short runs carry less signal).
- **Race rows** are pinned to $w = 1.0$ — ground-truth labels regardless of sensor dropouts.
- **Bonn HM (2024-04-14):** no HR strap $\Rightarrow$ HR features median-imputed from the 28-day pre-race window of other runs.
- All tabular models (GAM, RF, GBR, SVR) consume w via `sample_weight`; mean weight per LORO fold: 0.60–0.71.

Descriptive Summary









Commercial Race Predictors and Their Limitations

Platform	Method	Key Limitation
Riegel (1981)	Power law from one reference race	Ignores all training history
Strava	Riegel-like; updates on new race	Step function; no load sensitivity
Garmin	VO_2max estimate + power law	Proprietary; black-box

Shared limitations:

- **Race-only calibration** — hundreds of training runs discarded.
- **Flat-course assumption** — same prediction regardless of course profile.
- **No uncertainty quantification** — point estimate only, no prediction interval.
- **Opaque models** — no interpretability beyond the formula itself.

$\Rightarrow$ **Our approach:** use *all* training runs via pseudo-label inversion, transparent models with prediction intervals, and Minetti elevation correction.

Methodology

1. **Pseudo-label inversion**
   - Each training run provides an observed pace at a known effort level.
   $$\text{effort} = \frac{\text{HR}}{\text{HR}_{\text{max}}}$$
   - The model learns:
   $$\text{pace} = f(\text{distance}, \text{effort}, \text{fitness state}, \text{calendar})$$
2. **Temporal honesty**
   - For each race, the model only uses activities recorded **before** that race date.
   - This avoids using future training data to predict past races.
   - Errors can therefore be interpreted as realistic race-day prediction errors.

Methodology

3. **Leave-one-race-out cross-validation**
    - With six race labels, each race is held out once.
    - The model is trained on non-race activities recorded before the held-out race.
    - This gives one realistic prediction error per race.

Model Overview: GAM and Tree-Based Models

1. **Generalised Additive Model (GAM)**

$$\text{pace} = \beta_0 + s_1(\text{dist.}) + s_2(\overline{\text{HR}}) + s_3(\text{GAP speed}) + s_4(\text{TSB}) \\ + f(\text{dow}) + f(\text{month}) + \varepsilon$$

   - Smooth terms: natural cubic splines.
   - Regularisation selected by grid search.

2. **Random Forest and Gradient Boosting**
   - `RandomForestRegressor`: 400 trees.
   - `GradientBoostingRegressor`: 300 trees, max depth 3.
   - Additional quantile-loss boosters at $\alpha = 0.05$ and $\alpha = 0.95$ provide a 90% prediction interval.

Model Overview: SVR and Neural Network

3. **Support Vector Regression**

$$C \in \{1, 10, 100\}, \quad \gamma \in \{\text{scale}, 0.01, 0.1\}, \quad \varepsilon \in \{1, 5, 10\}$$

- RBF-kernel SVR inside a `StandardScaler` pipeline.
- Best full-table parameters: $C = 100$, $\varepsilon = 1$, $\gamma = 0.01$.

4. **Convolutional Neural Network**

$$X_i \in \mathbb{R}^{4\times T}, \quad \text{channels} = [\text{GAP}, \text{HR}, \text{cadence}, \text{grade}], \quad T = 7200$$

- Preserves the per-second sequence instead of collapsing runs to summary statistics.
- Three `Conv1d` blocks followed by pooling, dropout, and dense layers.
- Loss: MSE on log(pace); optimiser: AdamW, 30 epochs per fold.

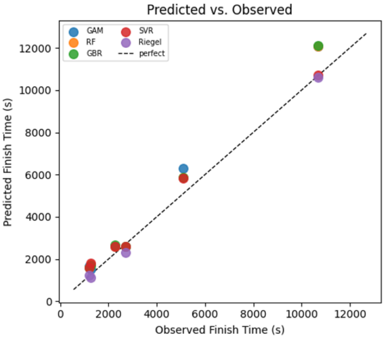
Results: Leave-One-Race-Out Prediction Errors

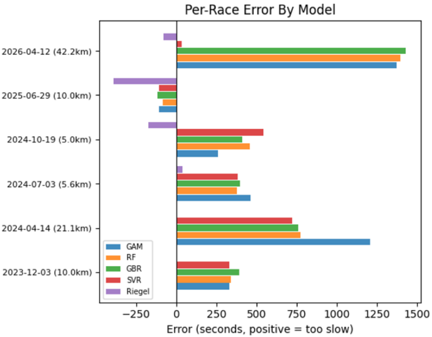
Signed prediction error in %. Positive = predicted slower than actual.

Race	Dist.	Actual	GAM	RF	GBR	SVR	NN
Köln Nikolauslauf	10 km	0:37:43	+14.6	+15.0	+17.4	+14.6	+27.1
Bonn HM†	21 km	1:24:53	+23.7	+15.2	+14.9	+14.2	−42.0
Lousberglauf	5.6 km	0:19:58	+38.7	+31.6	+33.0	+32.1	+13.0
DIY-Triathlon	5 km	0:21:18	+20.3	+35.7	+32.2	+42.5	−12.9
Schwarzwald Drachen.‡	10 km	0:44:57	−3.9	−3.2	−4.3	−4.0	+118.5
Milano Marathon	42 km	2:58:16	+12.8	+13.1	+13.4	**+0.3**	+3.0
MAPE			19.0	19.0	19.2	**18.0**	36.1

† No HR sensor; HR features imputed. ‡ Run after swim + bike; fatigue state outside training distribution.

Results: Model Comparison

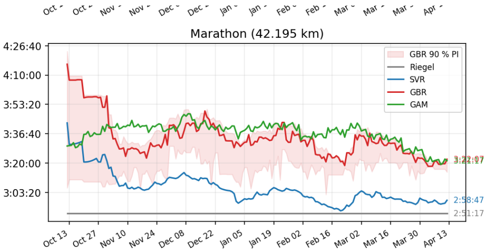




Left — Predicted vs. Observed: SVR sits on the identity line for the marathon; all others overshoot.

Right — Per-Race Error (s): Marathon: SVR ≈ 0 s; others $+1\,000$–$1\,400$ s. Short races: systematic over-prediction; triathlon: Riegel under-predicts.

Results: Marathon Predictor — Daily Estimates (6-Month View)



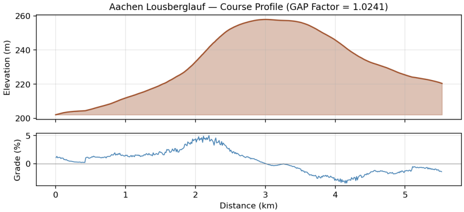
Riegel (grey flat line): $T_2 = T_1 \cdot (D_2/D_1)^{1.06}$ (Riegel, 1981). Reference (T_1, D_1) is unchanged between races $\Rightarrow$ constant prediction. Steps only when a new race is ingested; no sensitivity to daily training load. MAPE = 8.0% over 4 folds — best of all methods despite this limitation.

Model	Pred.	Err.
SVR	2:58:47	+0.3%
NN*	3:03:37	+3.0%
Riegel	2:56:59	−0.7%
GBR	3:22:07	+13.4%
GAM	3:21:08	+12.8%
Actual	**2:58:16**	—

- **SVR** (blue): 31 s off — best model.
- **GBR/GAM**: +13% over; PI narrows with build-up.
- **NN***: 2nd best (+3.0%); not in figure.

*From LORO evaluation only.

Results: Course Elevation — Lousberglauf (5.555 km)



Course: 5.53 km, ↑56 m / ↓37 m.
GAP factor = 1.0241 (Minetti, segment-weighted) ⇒ course is ∼2.4% harder than flat.

Model	Flat	Adj.	Δ
GAM	27:42	28:22	+40 s
RF	26:17	26:54	+38 s
GBR	26:33	27:12	+38 s
SVR	26:23	27:01	+38 s
NN	22:34	23:06	+33 s
Actual		**19:58**	—

- Models learn pace on **GAP** ⇒ flat-course predictions.
- Multiplicative course correction recovers terrain effect at inference time.
- **Riegel, Strava, Garmin**: one number per distance, no course input.
- Lousberglauf gentle (+2.4%); mountain races can exceed +15%.